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Effect of Inorganic Scales on Water Injection Performance: Quantitative Analysis, Mitigation, and Prevention Measures

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Abstract

In this work, we propose a practical and straightforward workflow to estimate time-dependent injectivity impairment caused by inorganic scale. The workflow integrates basic laboratory studies with analytical and numerical analysis of reservoir rock permeability reduction. The proposed method was applied to a planned water injection project in an oil-producing reservoir of a brownfield in Kazakhstan. For the project, we used a scale prediction simulator to identify the main type of expected precipitate (calcium carbonate). We also estimated trends for other scales precipitating (hydroxides) under different pH conditions, measured the actual scaling potential in the laboratory for both unmodified and pH-modified injection water, applied the Woods and Harker model to predict scaling distribution with distance, estimated the scale volume fraction distribution in pores over a three-year period, and calculated the resulting injectivity reduction. Conclusions were drawn regarding why matrix cleanup treatments are ineffective for mitigating injectivity impairment caused by scale precipitation.

Introduction

Definition and Formation Mechanisms of Inorganic Scales

For this study, we will define inorganic scales as solid deposits formed within reservoir rock and primarily composed of carbonates, sulfates, and sulfides. The most common types include calcium carbonate, calcium sulfate, barium sulfate, strontium sulfate, iron (II) carbonate, and iron sulfides. Key mechanisms that trigger the formation of inorganic scales include, but are not limited to (Shajari and Rashidi, 2021; Osode et al., 2015):

- Changes in the reservoir temperature and pressure,
- Variations in the pH and gas content (e.g., CO_2) of the formation fluid,
- Changes in the fluid's chemical composition due to the injection of water from the surface.

The Importance of the Inorganic Scale Precipitation Problem

The precipitation of inorganic scale in porous rock during water injection has been a well-known problem for decades in various locations worldwide (Vetter and Phillips, 1970; Hardy et al., 1994; Gomes et al., 2002; Xu et al., 2024). For water injection wells (e.g., during water flooding as a method of enhanced hydrocarbon recovery), the primary negative consequence of inorganic scale accumulation is permeability impairment, which leads to a decrease in the injectivity index. Key parameters controlling the severity and rate of reservoir rock plugging by inorganic scale include:

- Water injection rate,
- Ionic content and temperature of the injected water,
- Reservoir depletion rate (for scale triggered by changes in reservoir pressure).

Proper selection of these control parameters allows for the mitigation, if not prevention, of inorganic scale accumulation.

The Effect of Inorganic Scale on Injectivity Impairment: Studies of Particulate Plugging of Pore Space

Assuming that inorganic scale behaves as particulate matter while traveling through porous rock, we identified a family of studies that concentrate on the process of particulate plugging of pore space during injection.

Davidson (1979) used laboratory data to analyze the transport of silica particles through pore space with fluid. He derived a solution for the critical injection rate (CIR) for a fluid carrying particles through a porous medium. Below this rate, particles deposit on the rock surface, and deep particle invasion into the rock is unlikely to occur. According to Davidson, the CIR is directly proportional to the initial pore diameter and inversely proportional to the particle diameter. This author also derived solutions for permeability reduction from particulate bridging and the time required for this reduction to develop. For example, for a formation with 25% porosity, an 8-inch wellbore diameter, a particle size of 2 microns, and a pore size of 20 microns, the CIR is approximately 34 m^3/d per meter of wellbore length. If the injection rate under the same conditions is reduced to approximately 11 m^3/d per meter, a 50% permeability reduction will occur within a distance of 16 wellbore radii after 7 years.

Vetter et al. (1987) not only provided a thorough review of several laboratory and analytical studies on injectivity impairment from particle invasion but also conducted specific tests proving the importance of accounting for the role of sub-micron particles in the damage mechanism.

Pang and Sharma (1997) summarized key laboratory and analytical works on injectivity impairment due to particulate penetration. They suggested four types of curves for injectivity decline versus injected volume and proposed a model for injectivity decline that accounts for internal and external particle accumulation.

As reported by McDowell-Boyer et al. (1987), laboratory work on particle capture by porous media and its quantitative interpretation was performed by Sakthivadivel (1969), who found that the most significant permeability damage (by a factor of 7 to 15) and pore space occupation (30% and higher) occur when the ratio of the median grain diameter (d_m) to the particle diameter (d_p) is between 10 and 20. For small particles ($d_m/d_p > 20$), permeability reduction is limited to 10–50%, with only a 2–5% decrease in pore volume. For large particles ($d_m/d_p < 10$), the study revealed no particle invasion and the formation of a filter cake.

Inorganic Scales Precipitation: Complex Models

Another set of studies treats the inorganic scale precipitation process as a complex physicochemical phenomenon.

Woods and Harker (2003) considered the mixing of sulfate-rich injected water with barium-rich formation water in porous rock. They found that the numerical solution of the precipitation process can be represented by an analytical solution in which the mass of forming solids decays at a rate inversely proportional to the square root of the distance to the mixing front.

Woods (2003) also provided a quantitative estimation of particle dispersion due to the tortuous path they follow in pore space and the decorrelation of fluid trajectories on grain surface irregularities. An important implication of this phenomenon is that scale precipitation (triggered by the injected fluid's composition) may spread away from the apparent main injection stream.

Bedrikovetsky et al. (2009) developed an analytical model for injectivity decline and the size of the damaged zone during the injection of scale-prone water. One of their important findings is that precipitation which significantly affects well injectivity takes place in a zone extending 10–20 well radii from the wellbore. This distance can be taken into account when planning scale removal (e.g., acidizing) and scale inhibition operations.

Vetter et al. (1982) developed a scale prediction model and applied it to all possible mixing scenarios and ratios involving two injection waters and two reservoir waters. Their model also considered precipitation not only in the reservoir but also in the wellbore and surface equipment. Osode et al. (2015) used a similar analysis of various water mixture ratios, along with laboratory core-flow tests, to analyze the potential injectivity performance of different injection water types for a specific industrial project.

Project Overview, Problem Statement and Study Goals

A water flooding project (WFP) is planned for a brownfield located in Western Kazakhstan to increase the recovery factor and compensate for the decline in reservoir pressure. The water for injection is planned to be taken from layers above the target reservoir. The injection water and reservoir water contain a significant amount of ions (Appendix A). During preparation for the WFP, the first step is a study of the effect of inorganic scale on injectivity. The study has two goals:

1. To provide a general overview of the scaling tendency and, consequently, the risks of injectivity decline in the target reservoir for the given water sources.
2. To define the direction for further, more detailed (time- and investment-demanding) studies.

Based on the stated goals, the study has the following requirements and constraints: it must be time- and cost-efficient, use basic laboratory tests (e.g., water analysis without core-flow tests), rely on an analysis of industry experience, and employ simplified numerical and analytical calculations (i.e., without full-field simulations).

This paper outlines the main methods, results, and conclusions from the inorganic scaling study performed under these requirements and constraints.

The Effect of Inorganic Scales on Injectivity Impairment: a Case Study

Target Reservoir Properties

Characteristics of the reservoir and field where WFP is planned are provided in Table 1.

Table 1—Information about the target reservoir.

Parameter	
General	
Average reservoir TVD, m	~3000 m
Reservoir type	Porous, Sandstone, Shale Barriers
Hydrocarbon type	Oil
Field development method	Artificial lift (ESP)
Average net height, m	25-30
Average reservoir porosity, fraction	12-16
Permeability, mD	2-5
Reservoir temperature, °C	~120
Initial reservoir pressure, atm	~280-300
Current reservoir pressure, atm	~190-220
Current water cut, %	20-40
Injection wells for the water-flooding project	
Expected water injection rate per well, m^3/d	50-150
Completion type	Perforated casing
Injection water source	Water wells, from isolated layers far above from the target reservoir

Proof of Scaling Issue and Scale Type Identification: Comparative Dissolution Method

Scale accumulation in downhole equipment and the near-wellbore zone is one of the key challenges in this oilfield. The inorganic nature of the scale for this project we confirmed via a simple yet effective method: the comparative dissolution of actual scale samples in various fluids.

This method involves dividing one scale sample into several smaller samples of equal weight and relatively similar form (to achieve a similar surface area). Each small sample is then submerged in a specific fluid. By comparing the dissolved mass of the initially identical scale samples in various fluids, conclusions about the scale's composition can be made. The comparative dissolution method allows for the qualitative differentiation between organic and inorganic scale types, and a quantitative estimation of scale content (with a certain margin of error). This method is:

- Low-cost, as it requires only basic laboratory equipment (scales, bottles, and common fluids);
- Useful when advanced laboratory studies are not possible due to time and cost constraints.

Table 2 shows the results of the comparative dissolution method applied to actual scale samples from this oilfield.

Table 2—Scale type identification using the comparative dissolution method.

Dissolution fluid	Dissolution fluid purpose	Dissolved %	
		Scale Sample 1 From the tubing bottom close to perforations after tubing extraction	Scale Sample 2 From the Electric submersible pump surface after pump extraction for repair
Hydrochloric acid (15%)	Estimate amount of calcium and magnesium carbonate	91	50
Acetic acid	Estimate amount of ferric hydroxide – $Fe(OH)_3$ (which is poorly soluble in HCl)	79	42
Xylene	Estimate amount of asphaltenes (poorly dissolved in methanol)	3	15
Methanol	Estimate amount of naphtenate-salts containing scales e.g. calcium, sodium naphtenate (which is poorly dissolved in xylene)	13	7
Mutual Solvent	Estimate amount of paraffins, i.e. long-chain alkanes, and waxes (both poorly dissolved in methanol)	13	10
Diesel	Estimate amount of paraffins (which is soluble in diesel, but poorly soluble in methanol)	0	5
Qualitative and approximate quantitative conclusion on scale type:		~85-90% of calcium carbonate (with minor magnesium carbonate because of low magnesium content in reservoir water) ~10-15% of calcium naphthenate (with minor sodium naphthenate because of lower sodium content in reservoir water)	~45-55% of calcium carbonate (with minor magnesium carbonate) ~5% of napthenic acids/ resins ~5-10% of asphaltenes ~35-45% - reservoir particles of sandstone/quartz

Injection and Reservoir Water Scaling Tendency Classification

First of all, the scaling tendency of the injection and reservoir waters was confirmed using the Langelier Saturation Index, *LSI* (Langelier 1936) and Ryznar Stability Index, *RSI* (Ryznar 1944):

$$LSI = pH - pH_S \quad \text{(Eq.1)}$$

$$RSI = 2pH_S - pH \quad \text{(Eq.2)}$$

Where pH represents the measured pH of the water, while pH_s is the theoretical saturation pH for calcium carbonate, which is calculated from the water's ionic composition (refer to Appendix A).

With pH_s value ~4.3 for the injection water and ~3.9 for the reservoir water, both are classified as having strong scaling tendency (Table 3).

Table 3—Injection and reservoir water scaling tendency assessment.

Water	Langelier Saturation Index		Ryznar Stability Index	
	Value	Comment	Value	Comment
Injection	1.1	Scaling tendency water	3.2	Heavy Scale tendency water
Reservoir	2.1		1.8	

It is noteworthy that the target reservoir fluid contains CO_2, which has a complex effect on precipitation that depends on pressure and temperature conditions. These CO_2-triggered precipitation processes are not considered in this paper, as they require a separate study including laboratory tests and numerical simulation, as shown in industry literature (e.g., Yuan et al., 2021).

Simulations to Analyze Scales Precipitation Type and Trends

During the injection of water into the reservoir, many parameters change with time and distance as the fluid front propagates. The three key parameters for scaling tendency analysis are (i) the injected fluid to reservoir fluid ratio (IRR), (ii) reservoir temperature, and (iii) reservoir pressure.

The term "injected fluid front" refers to the distance from the injection well to the zone where only the original reservoir fluid exists. A transition zone always exists between the fluid front position and the zone fully swept by the injected fluid (accounting for residual saturations due to capillary forces). In this transition zone, the three parameters mentioned above are complex functions of reservoir and fluid properties and the injection rate. Therefore, the evolution of pressure and temperature should be estimated via hydrodynamic modeling, which is out of scope for this study. Consequently, for the estimation of pressure and temperature change over time, we will assume a linear evolution on the scaling tendency plots.

Based on the ion content given in Appendix A, the overall scaling potential for the injection water was estimated using scale prediction software. The approximate pressure-temperature (PT) evolution over a medium-term period of approximately 3–5 years was analyzed for two regions. Region 1 is in the vicinity of the injection well (from a few meters to a few tens of meters away). Region 2 is farther into the reservoir, approximately a hundred meters away from the injection well. Figure 1 shows the results of this estimation.

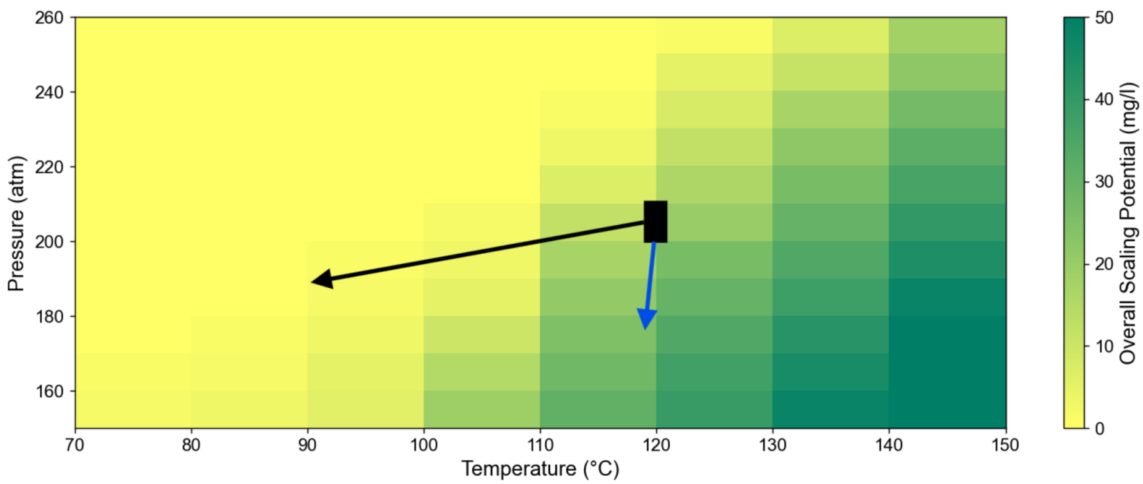

Figure 1—This figure shows the estimation of the overall scaling tendency of the injection water and tracks the evolution of possible pressure-temperature (PT) conditions over the medium-term (3–5 years) in two regions: near the injector well (black arrow) and in a region far from the injector well (blue arrow). The black rectangle indicates the most probable pressure-temperature (PT) conditions in the reservoir expected at the start of injection. The main observation from this plot is that formation cooling leads to a decrease in scaling risk within the vicinity of the injector well (even at declining pressure). However, in the far region, the scaling potential may increase over time due to inevitable reservoir depletion and the weak effect of cooling.

As mentioned earlier, in complex reservoir flooding processes, inorganic scales precipitation is a function of pressure, temperature and the chemical composition of the fluid. While the influence of the first two parameters (PT conditions) is well understood from plots like Fig. 1, the effect of water composition is better understood via the plots shown in Fig. 2.

In our study, the water composition gradually changes away from the wellbore as a result of the injected water mixing with reservoir water. Plots of scaling potential versus the ratio of mixed waters, generated by scale-prediction software for various PT conditions, allow for an understanding of major scaling tendencies. In our case, as shown in Fig. 2, three main conclusions can be drawn:

- For calcium carbonate, under the PT conditions at the start of the flooding project, the highest scaling potential occurs within an injection-to-reservoir water ratio range of 0.2 to 0.4. This means the maximum scaling tendency is not in the near-wellbore area (where the ratio is approximately 0.8, due to non-ideal sweep efficiency) nor at the fluid front (where the ratio is ~0), but somewhere in between.
- For calcium carbonate, the highest scaling potential remains within the injection-to-reservoir water ratio range of 0.2 to 0.4 for the entire range of PT conditions expected over a 3–5 year period.
- Various inorganic scales have different mixed water ratio ranges for their highest scaling potential.

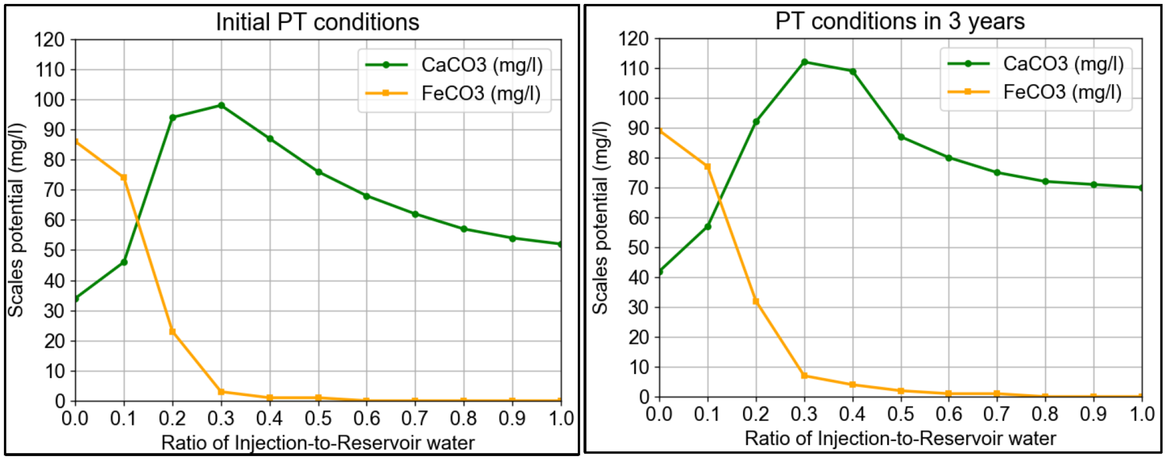

Figure 2—Scaling potential of the injection and reservoir water mixtures at different ratios for two inorganic scales: CaCO3 and FeCO3. The initial PT conditions (at the start of the water flooding project) are a reservoir temperature of approximately 120 °C and a reservoir pressure of 200 atm in the near-wellbore zone. The PT conditions used for simulations—a reservoir temperature of approximately 115 °C and a reservoir pressure of 170 atm—represent the expected reservoir conditions after 3–5 years of injection at greater depths (more than 30 m from the wellbore).

Scale Precipitation Prevention and Mitigation Options

Using scale precipitation prediction software, we studied the following water pre-injection treatment options as scale control methods.

Option 1: Increase of injection water pH with alkaline substances, followed by solids separation.

The idea behind this option is that as pH increases (due to the addition of alkaline substances), Ca^{2+} and HCO_3^- ions form $CaCO_3$ scale due to an equilibrium shift. By filtering out this scale on the surface before injection, the $CaCO_3$ scaling potential in the rock is decreased. However, if the pH of the injected water remains >9.5 after pre-conditioning, metal ions tend to form hydroxide precipitates. In our case, the reservoir water contains a significant amount of iron and magnesium, which would produce an excessive amount of precipitates after contacting the high-pH injection water (Fig. 3, left). Consequently, instead of calcium carbonate scale, an even larger amount of hydroxide scale is formed.

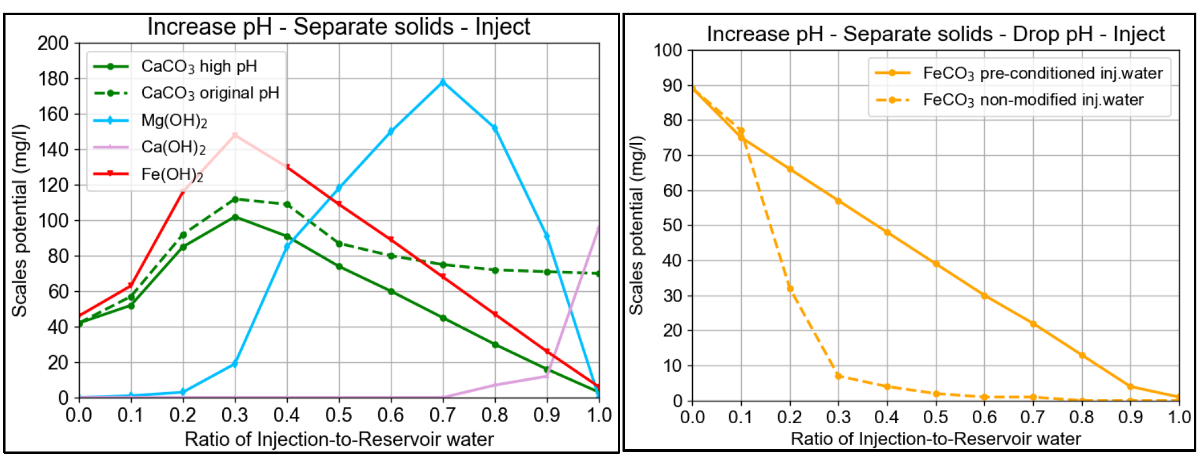

Figure 3—Scale potential variation for mitigation options employing pH preconditioning of the injected water. Option 1 (left plot) involves increasing the pH with specific chemicals, removing the precipitated solids at the surface, and subsequently injecting the water. Option 2 (right plot) suggests decreasing the pH to neutral values after solids separation at the surface. The PT conditions used for simulations—a reservoir temperature of approximately 115 °C and a reservoir pressure of 170 atm—represent the expected reservoir conditions after 3–5 years of injection at greater depths within the reservoir (more than 30 m from the wellbore).

Option 2: To avoid hydroxide precipitation, Option 1 can be improved by decreasing the injection water pH to near-neutral values after solids filtration and before injection.

In this case, the scaling tendency drops significantly for all scales except $FeCO_3$. For $FeCO_3$, the scaling tendency increases slightly at higher injection water concentrations (i.e., closer to the wellbore), as shown in Fig. 3 (right).

Option 3. This option involves heating the injection water at the surface to cause scale precipitation, followed by filtration of the precipitates before water injection.

Simulations for our case show that this method reduces the calcium carbonate scaling potential and even eliminates it in the near-wellbore zone (Fig. 4). This option does not reduce the scaling potential of calcium carbonate to almost zero, as Option 2 does. However, from an operational and investment point of view, preconditioning injection water by heating may be more attractive than a two-step chemical preparation process (pH increase followed by a pH drop).

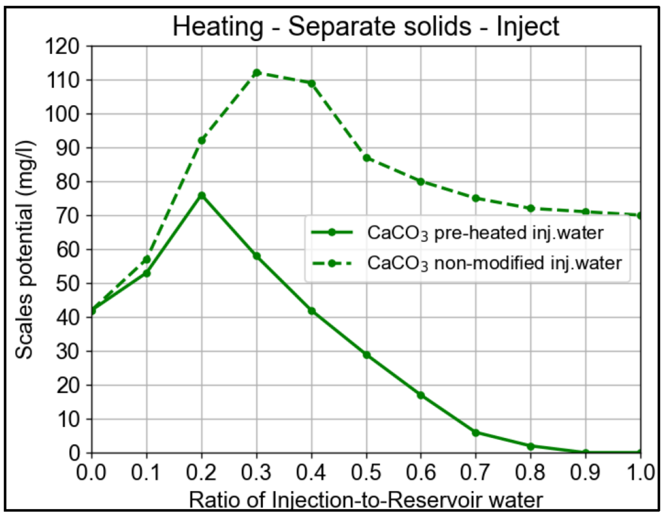

Figure 4—Scale potential variation for mitigation option employing heating pre-conditioning of the injected water. The PT conditions used for simulations—a reservoir temperature of approximately 115 °C and a reservoir pressure of 170 atm—represent the expected reservoir conditions after 3–5 years of injection at greater depths within the reservoir (more than 30 m from the wellbore).

Laboratory Tests to Quantify Precipitated Scale Potential

After simulations revealed the expected scale precipitation trends and the efficiency of mitigation options (Fig. 1 – Fig. 4), relatively simple laboratory tests were performed to quantify scale precipitation for a further estimation of its effect on formation permeability.

The laboratory tests involved heating water samples in a standard heater without pressurization (up to 93–95°C in closed bottles) for an extended period of time (~72 hours). Samples were selected to represent various conditions: 100% injection water (near-wellbore), 100% reservoir water (injected fluid front), and 50/50 and 60/40 mixes of injection and reservoir water (conditions approximately in the middle of the flooded zone). Additionally, to evaluate the pH-variation scale mitigation options discussed earlier, four variations of the 60/40 mix were tested: pH~5.6 (non-modified mix), pH=9.5, pH=10.5, and pH=6 (modified by a preliminary increase to 10.5, scale filtration, and then a decrease to 6). After heating, the precipitate was filtered, dried, and weighed. These test conditions should adequately represent the scale precipitation potential under reservoir conditions, based on the scale potential analysis software (Fig. 5).

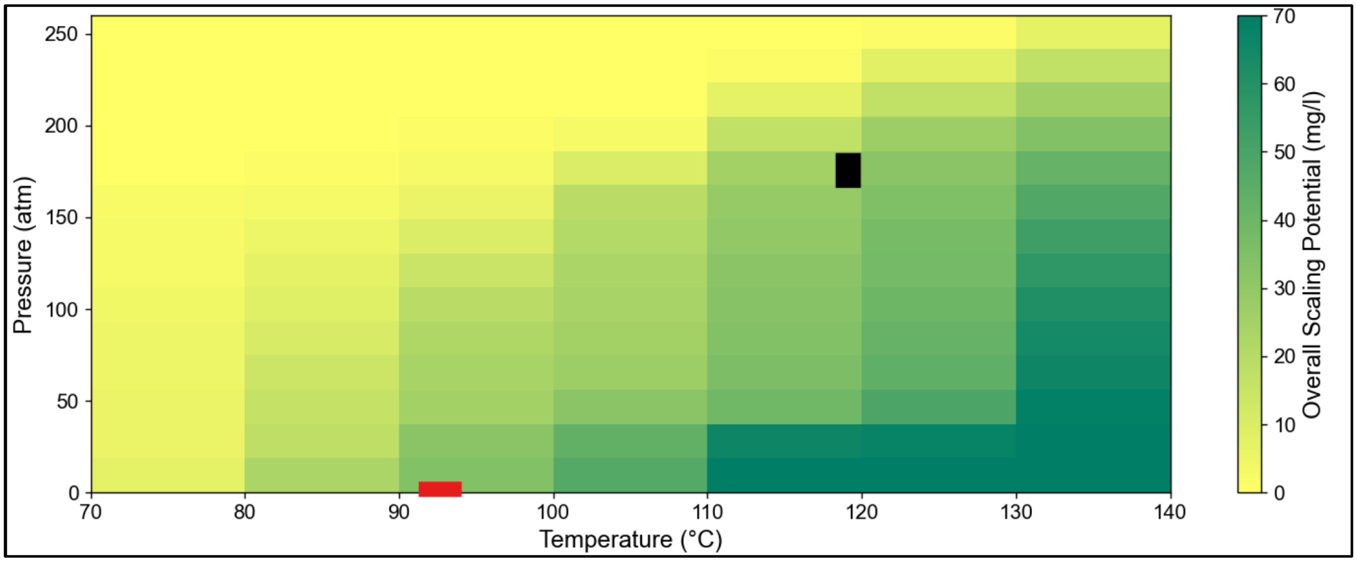

Figure 5—Estimation of the overall scaling tendency of the injection water. The black rectangle shows the most probable PT conditions in the reservoir expected between the beginning of injection and 3 years of injection, at a distance of ~30 m from the wellbore. The red rectangle represents the PT conditions of the laboratory tests in the heater without pressure. It can be concluded that the scale potential estimation in the simple laboratory test should satisfactorily represent the scaling tendency under the given reservoir conditions.

The results of the laboratory precipitation tests are shown in Table 4. Two key conclusions can be drawn from these results:

Table 4—Results of laboratory precipitation tests in the heater.

Sample	Precipitation (mg/l)	Comment
Injection water (IW)	2210	>70% dissolved in 15% HCl, possibility of ~30% of contamination not related to scaling (e.g. rust)
Reservoir water (RW)	3420	
50% IW + 50% RW	2730	
60% IW + 40% RW (non-modified pH~5.6)	2570	
60% IW + 40% RW (pH increase to ~9.5)	3830	~50% increase from non-modified mixture
60% IW + 40% RW (pH increase to ~10.5)	5360	~110% increase from non-modified mixture
60% IW + 40% RW (pH increased to 10.5, scale filtrated, then pH decreased to 6)	Traces (<10)	Almost complete elimination of precipitation at the test temperature

1. The weight of the precipitated scale is one order of magnitude higher than the weight predicted by the scale prediction software. Possibly, some of the precipitation observed in the laboratory may be attributed to contamination from the vessels and tubing where the water was stored. However, the difference is much higher than the potential contribution from contamination. Consequently, the scale prediction software should be used for qualitative and comparative analysis only: to assess trends for different scale types and to evaluate scale mitigation scenarios.
2. The laboratory tests confirmed the scale prediction software's result that scale mitigation Option 2 (increase pH → filter precipitate → decrease pH) eliminates almost all potential calcium carbonate scale.

During the laboratory tests, the evolution of scale precipitation over time was clearly observed (Fig. 6). Dissolution tests in 15% hydrochloric acid indicated greater than 70% solubility, suggesting that the majority of the scale is calcium carbonate. However, up to 30% contamination may be present in the precipitated mass. Consequently, the precipitation mass from Table 4 should be multiplied by approximately 0.7 for further quantitative analysis (e.g., the injection water precipitation potential decreases from 2210 to ~1550 mg/L, the reservoir water potential becomes ~2400 mg/L, with an average between both waters of approximately 2 g/L).

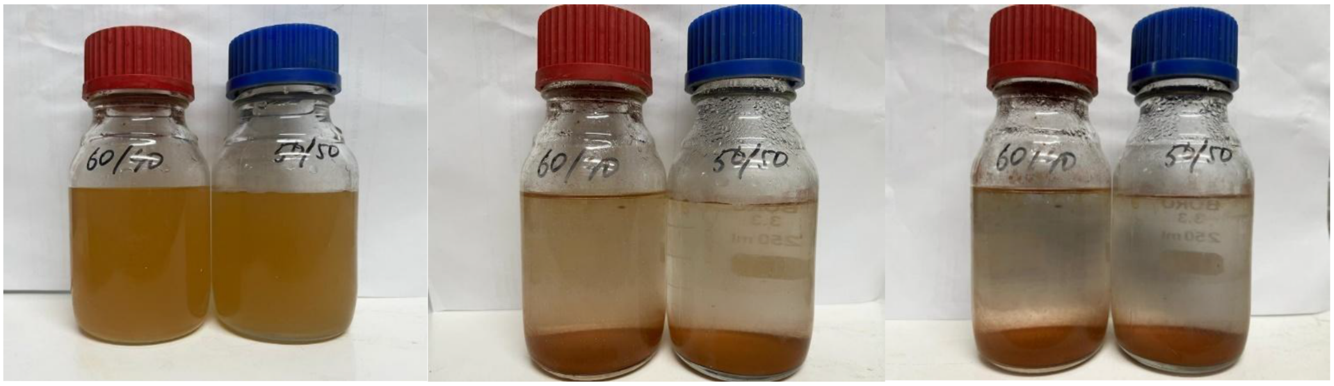

Figure 6—Scale precipitation development during heating test for two mixtures of injection and reservoir water (ratios 60/40 and 50/50, respectively).

Effect of Inorganic Scales Precipitation on Injectivity Impairment

The results of laboratory tests and scale prediction software simulations are used for a quantitative analysis of permeability reduction and injectivity impairment in the given field.

Industry literature review. The effect of permeability reduction due to scale accumulation is described in numerous publications. Ohen and Civan (1993) provided a detailed model for calculating permeability decrease based on scale accumulation processes. Santos et al. (2017) studied how accumulation in the porous medium reduces absolute permeability and thus productivity. Standard thermodynamic models may not be sufficient to describe the scaling problem due to the solid phase in the solution; precipitation depends on specific phenomena such as the type of flow and the adherence of particles to the rock face. Zhang and Farquhar (2001) determined that the rate of calcium carbonate scaling increases with the injection rate. However, it is reasonable to expect that at higher flow rates, plugging may take longer to occur because of particle transport. Thus, the impact of the injection rate on scale deposition depends on the balance between increasing scale mass at a higher rate (a higher volume of scale source per unit of time) and delayed plugging due to drag forces. In addition, the process is complicated by the practically undetectable type of permeability impairment, which can be pore filling with filter cake, pore throat bridging, or particle migration. Overall, the complexity of the parameters involved in the permeability impairment process requires certain assumptions to be made for quantitative modeling.

Regarding the quantitative estimation of permeability impairment, industry data varies significantly with reservoir properties and scale type. Moghadasi et al. (2002) reported that scales (sulfate, in their case) result in a fourfold injection rate drop over five years in field conditions. Binmerdhah and Yasin (2009) performed a core flooding test on sandstone with a porosity of ~30% and a permeability of ~10–15 mD. Although the main scale type expected in their test differed (calcium sulfate vs. calcium carbonate in our case), the permeability range matches our case. The higher porosity in their study, compared to our conditions, provides a basis for estimating an upper bound for permeability reduction. According to their study, at 80°C, scales may cause formation permeability to drop below 75% of the initial value after just two hours under core flow test conditions, which could equate to within a few days of actual injection.

Santos et al. (2017) reported results of a core flow study on Berea sandstone where the retained permeability can be as low as 15–20% after only 18 injected pore volumes. A test performed by Al-Taq et al. (2017) found a permeability reduction range of up to 20% for a 1.3-mD core with ~100 mg/L of total suspended solids (TSS) in the flooding fluid, and a 10% reduction for a similar TSS concentration. On the other hand, field trials performed by the same authors show that laboratory permeability reductions are significantly underestimated. In the field, the permeability impairment resulted in a significant injection rate reduction: after cleanup with acid, the reservoir showed an injectivity index increase by a factor of 3 to 8, corresponding to an apparent permeability improvement by the same magnitude. However, the duration of the effect from such matrix cleanup treatments depends significantly on the depth of scale precipitation around the well: if scales are expected to precipitate deeper than 5 meters from the wellbore, matrix treatments (which have a radius of penetration of less than 3 meters) will have only a short-term effect.

Simplified hydrodynamic modelling to estimate scale accumulation effect. To estimate the scale accumulation phenomena for our case, a simplified hydrodynamic simulation was performed with several assumptions.

An injector-producer well pair was placed 270 meters apart, with a 10-meter cell mesh between them. Reservoir properties were selected based on the target interval data (refer to Table 1). The scale precipitation tests (Table 4) showed that approximately 2 grams precipitate from each liter of fluid. Assuming a precipitate density of 2.7 g/mL (which is similar to carbonate scale), the average volume of precipitate per liter is ~0.74 mL, resulting in a volumetric concentration of potential scale in the injected fluid (VCS) of ~0.00074 v/v. The scale contained at this concentration does not precipitate uniformly as the fluid moves through the reservoir. Woods and Harker (2003) showed analytically and numerically that precipitation is inversely proportional to the square root of the distance from the injection point. Knowing the expected injection rate per well (Table 1, average 100 m^3/d), the effective porosity of the reservoir that injected water can fill (~12%), and the thickness of the target flooded layer (~7–10 m), it is possible to estimate that the fluid front will propagate to a distance of ~110 m in one year of non-stop injection (or 1.5 years considering inevitable injection interruptions due to operational procedures and maintenance in such projects). Based on the total injected volume within this period, the total precipitation mass distributed within this propagation radius can be estimated at ~60 to 80 tons. Applying the Woods and Harker solution, this scale mass distribution along the propagation distance can be illustrated (Fig. 7).

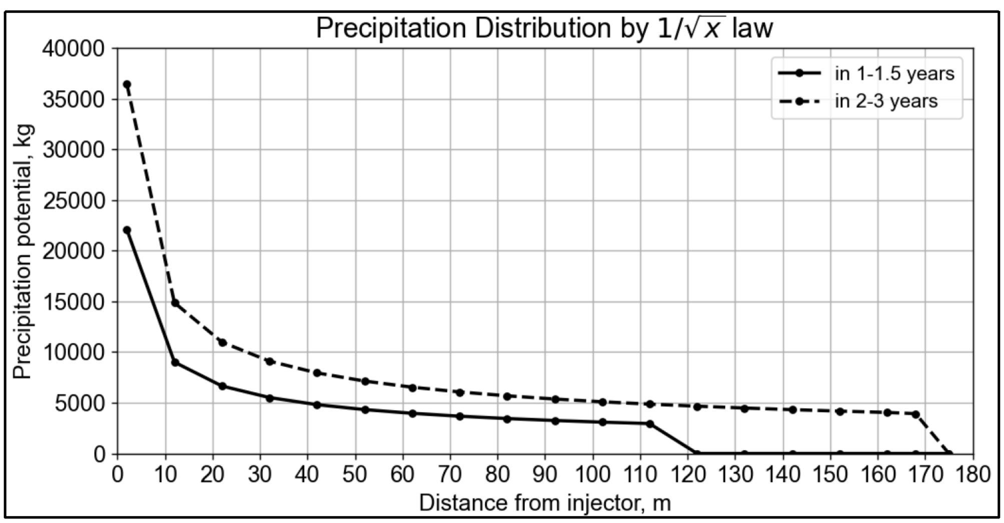

Figure 7—Scale precipitation distribution for the given project after approximately 1.5 and 3 years of injection. The sum of all points on each curve is equal to the mass of precipitate contained in the volume of water pumped during the given period. The distribution shown assumes that the injector (at zero on the x-axis) is the point where the precipitation process is most active. In reality, the location of the scaling initiation point depends on the scale type and PT conditions, as shown in Fig. 2 and Fig. 3.

From Fig.7 we extract a parameter called the precipitation intensity multiplier *PIM*:

$$PIM = \frac{PP_{inj}}{PP_{stab}} \quad \text{(Eq.3)}$$

where PP_{inj} is the precipitation potential in the vicinity of the "injector", i.e. near the point where precipitation process is the most active, PP_{stab} is the average precipitation potential far from the "injector", where precipitation potential flattens out.

Transition from the mass of solids into permeability is performed by estimating the Solid Volume Fraction (SVF) in the reservoir pores:

$$SVF = \frac{V_{sol}}{V_{por}} \tag{Eq.4}$$

Where V_{sol} is the volume of solids accumulated in the rock pores, V_{por} is a total pore volume, both parameters are defined for a given zone of the reservoir.

V_{sol} is a complex function of VCS, PIM, reservoir porosity, injection rate, and PT conditions, this is why numerical hydrodynamic simulations are required to estimate the SVF distribution.

Based on industry data on permeability reduction from precipitates (discussed earlier), we assume that permeability reduction depends on SVF as shown in Fig.8.

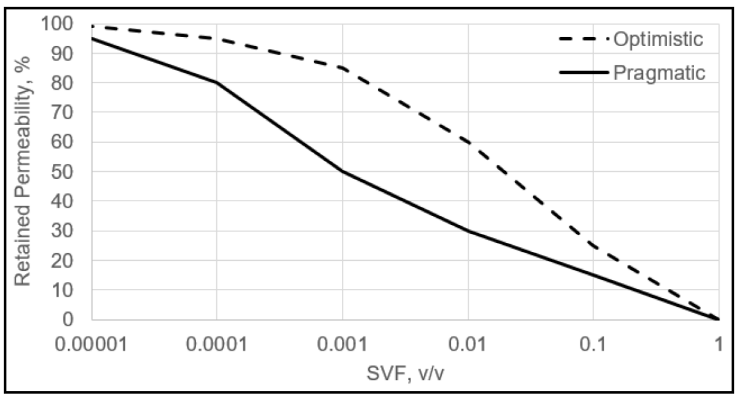

Figure 8—Retained permeability decrease with increase of SVF, for Optimistic and Pragmatic options.

Using all this information as input for the hydrodynamic simulator, we obtained the relative change in the Injectivity Index for both pragmatic and optimistic approaches for the base case precipitation potential of 2 g/L (Fig. 9, left). Furthermore, we estimated the effect of a possible decrease in precipitation potential for the optimistic approach: 2 g/L for the base case, 1.5 g/L for the first decrease option, and 1 g/L for the second decrease option (Fig. 9, right).

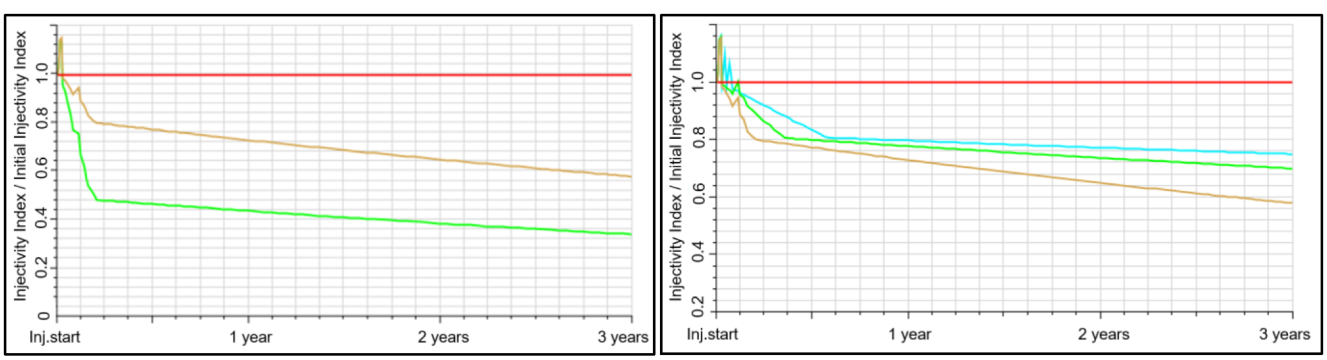

Figure 9—Injectivity impairment analysis from hydrodynamic simulations. Left: Results for the base case precipitation potential of 2 g/L, showing optimistic (brown) and pragmatic (light green) approaches for the permeability reduction versus solid volume fraction correlation. Right: Sensitivity analysis of the optimistic approach for the permeability reduction versus solid volume fraction correlation, for various precipitation potentials: brown for the base case (2 g/L), light green for 1.5 g/L, and blue for 1 g/L.

A decrease in precipitation potential could occur if a softer water source (with fewer scaling ions) is found for the project, or if a water pre-conditioning method—such as moderate heating and filtration—is approved. It can be seen from the right plot in Fig. 9 that a decrease in precipitation potential to 1 g/L and 1.5 g/L has a positive impact, especially in the longer term (resulting in a 12 to 16% increase in the injectivity index compared to the base case). However, only the use of scaling mitigation Option 2 (increase pH → filter precipitate → decrease pH) can provide the most effective preservation of the injectivity index, as only traces of scale were observed in the lab tests after the application of this option (Table 4).

Injection fluid front propagation estimated analytically (Fig.7) is very close to hydrodynamic simulations (Fig.10).

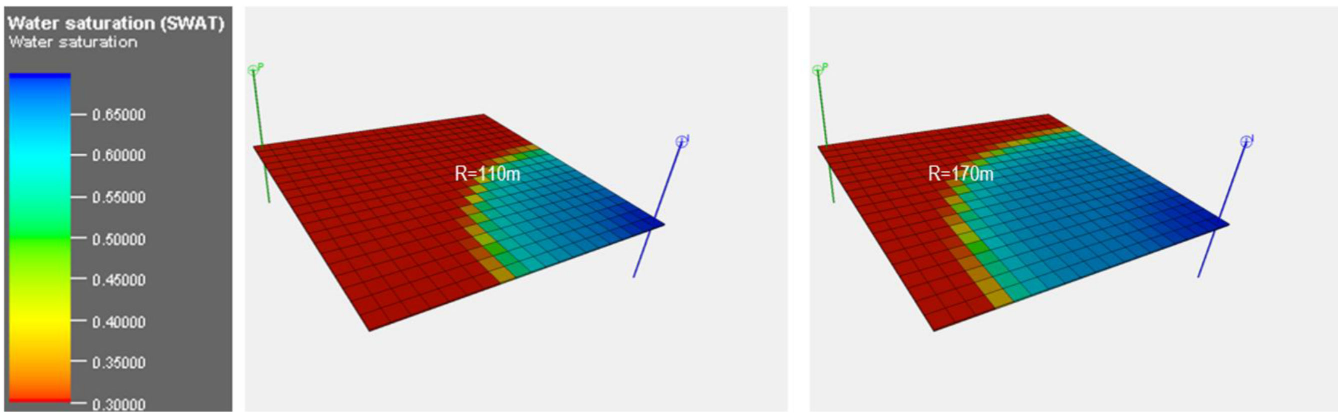

Figure 10—Fluid front propagation at 1 year (left) and 2 years (right), equal to approximately 110 m. and 170 m. respectively.

The estimated SVF distribution at 1, 2, and 3 years after the start of injection is shown in Fig.11.

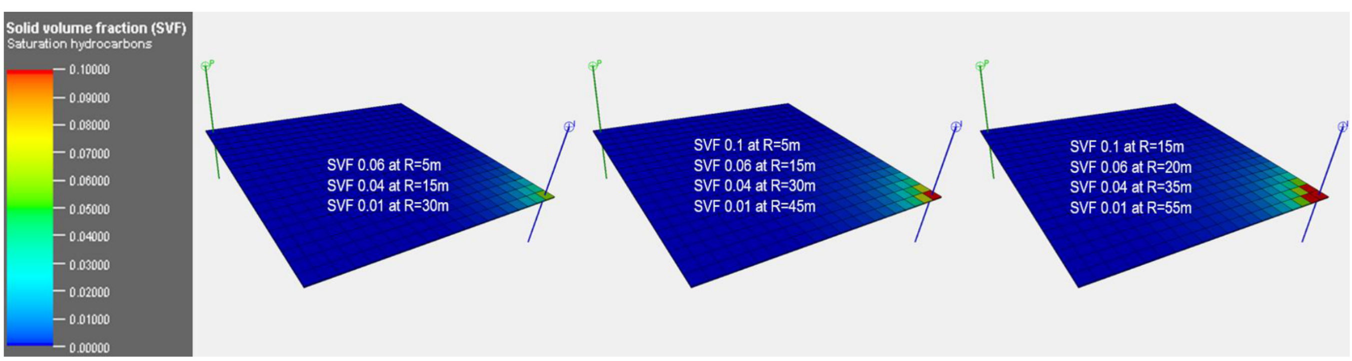

Figure 11—SVF at 1, 2, and 3 years after the beginning of injection.

An important conclusion from Fig. 11 above is that within one year, scale precipitation above 0.01 SVF extends 30 meters into the reservoir. This is one order of magnitude larger than the penetration radius of any scale-dissolving agent pumped in matrix mode without fracturing, such as HCl or other acids (< 3 meters). After three years, precipitation occurs as deep as 55–60 meters into the reservoir, with critical matrix damage reaching an SVF of 0.1. This leads to the conclusion that any matrix cleanup treatment involving acid injection will only provide a short-term improvement in water injectivity. Furthermore, there is a chance that after acidizing treatments, the damaged zone will propagate even deeper into the reservoir. This is because a temporary decrease in skin factor in the critical matrix zone will allow the damaging injection water to penetrate further.

Consequently, matrix cleanup or acidizing treatments are not a primary solution for addressing this scaling issue. Instead, pre-conditioning of the water through pH modification, heating, and filtration should be performed to decrease its scaling potential.

Conclusion

- The Comparative Dissolution Method allows for cost- and time-efficient determination of scale type in the laboratory through simple dissolution tests in specifically selected fluids. The method is based on a fit-for-purpose selection of fluids. For this case study, the use of hydrochloric acid, xylene, methanol, diesel, and an organic solvent allowed for distinguishing between inorganic scales, organic scales, and reservoir particles, and for estimating the percentage of each scale type with approximately 5% precision.
- An inorganic scaling simulator allows for the qualitative determination of scaling trends: the types of scales precipitated at given pH and PT conditions. For the considered case study, the prevailing scale type for the injection water is calcium carbonate.
- An inorganic scaling simulator allows for the qualitative analysis of scaling evolution when various scale control methods are planned. For this case study, an increase in injection water pH above 10, followed by the removal of precipitated solids at the surface and water injection, results in the activation of severe hydroxide precipitation. However, if the pH is reduced after precipitate removal (before water is injected into the reservoir), the scaling tendency for calcium carbonate decreases significantly.
- A simple scale precipitation test at elevated temperature for an extended time (48–72 hr) showed that the scaling potential is much higher in reality than predicted by the scaling simulator. Consequently, simple precipitation tests are more robust for the quantitative analysis of scaling potential than modeling, due to the multiple assumptions and unknowns used in simulators. For this case study, the scaling simulator predicted a maximum of approximately 120 mg/L of calcium carbonate scale for the injection and reservoir water mixture, whereas the scale precipitation test at ~95°C showed an accumulation of approximately 1550 mg/L of acid-soluble carbonate scale.
- The results of scale precipitation tests represent the volumetric concentration of potential scale in the injected fluid (VCS). The VCS is distributed according to the Woods and Harker model of precipitation versus distance (inverse square root). For our case study, this model results in a ratio of scale mass in the near-wellbore to scale mass at the fluid front (PIM) of 7:1.
- At any distance from the injector well, the Solid scale Volume Fraction (SVF) in the pores can be calculated using a hydrodynamic simulator as a function of VCS, PIM, injection rate, and reservoir properties. For this case, simulation results showed that in 1 year, the 0.01 SVF zone extends 30 meters into the reservoir; in 3 years, this zone extends approximately 60 meters into the reservoir, with critical matrix damage reaching an SVF of 0.1.
- Using SVF and an assumed function of permeability decrease versus SVF, injectivity impairment over time can be estimated in a hydrodynamic simulator. For this case, the injectivity index is expected to drop by approximately 25–55% in 1 year and by 40–65% in 3 years if non-modified injection water (~2 g/L) is used. The use of preconditioned injection water with half the scale potential (~1 g/L) results in a 16% higher injectivity index after 3 years compared to non-modified water.
- Matrix injection treatments (cleanup, acidizing) are not an effective way to mitigate the scale problem due to the extension of the high SVF (>0.1) zone far beyond the effective radius of matrix treatments. Water preconditioning through pH modification, heating, and filtration is the most effective way to prevent and mitigate scaling.

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Appendix A

Ion content of the injection and reservoir water:

Parameter	Injection Water	Reservoir water, well A	Reservoir water, well B
Specific gravity	1.1	1.1	1.09
pH	5.38	5.89	5.92
Ions (mg/l):			
Cl^-	48000	94000	90000
SO_4^{2-}	100	200	200
Ca^{2+}	33000	48000	47000
Mg^{2+}	<20	1000	2000
Fe	<12	300	280
CO_3^{2-}	<2	<2	<2
HCO_3^-	980	180	140
Hydroxides	<1	<1	<1